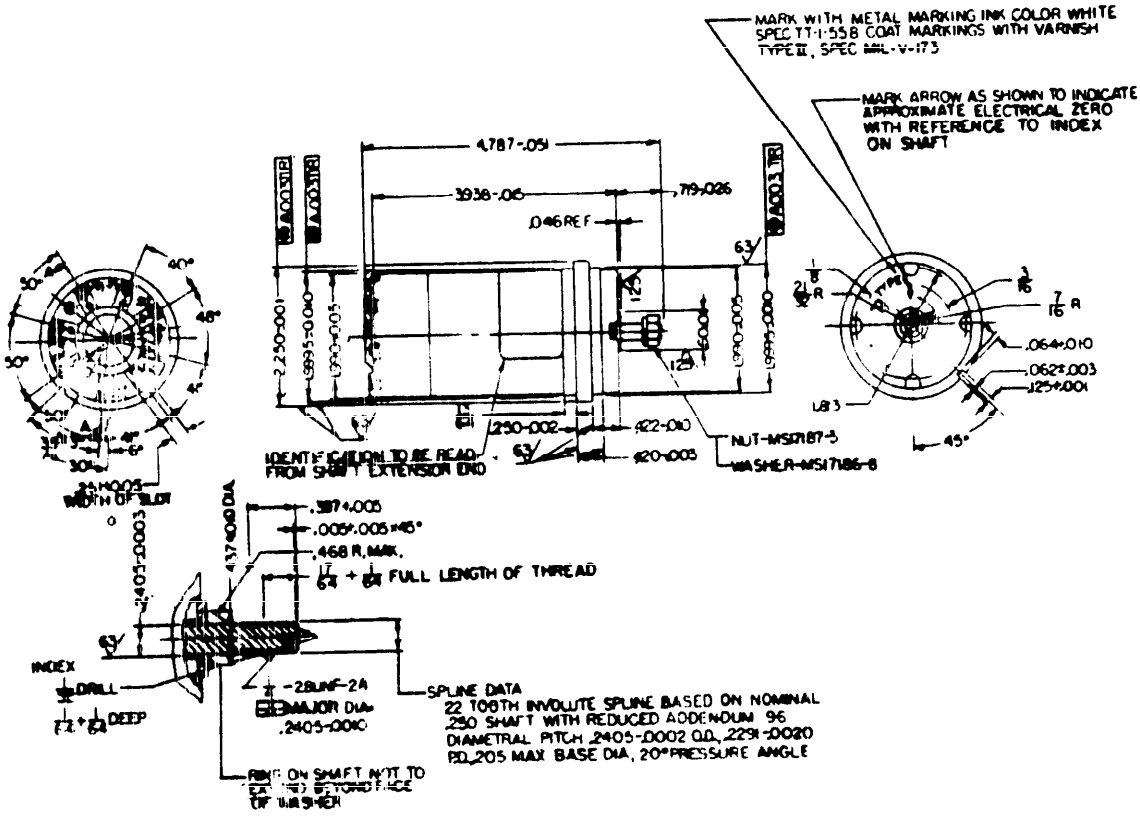
MIL-R-23417/16A(MU)
17 November 1969
SUPERSEDING
MIL-R-23417/16
21 December 1964

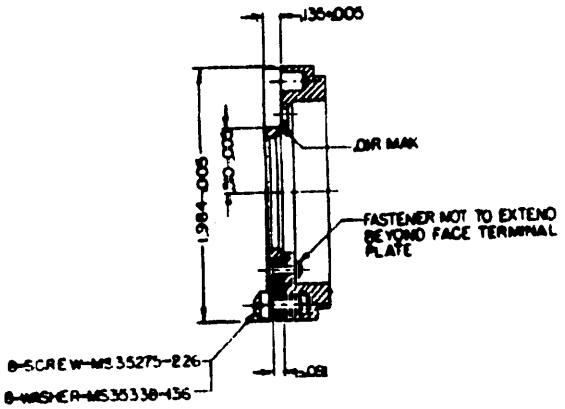
MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL: TYPE 23R32N4A

The complete requirements for procuring the resolver described herein shall consist of this document and the issue in effect of Specification MIL-R-23417.



PARTIAL SECTION SHOWING SPLINE SHAFT EXTENSION



PARTIAL SECTION A-A SHOWING TERMINAL BOARD
SCALE 2/1

NOTE: THIS SPECIFICATION IS TO BE USED TO SUPPORT IN-SERVICE MATERIEL ONLY AND NOT TO BE USED FOR NEW DESIGN.

NOTES:
1-ECCENTRICITY OF SHAFT DIAMETER (SMOOTH PORTION) SHALL NOT EXCEED .001 TIR WHEN SHAFT IS ROTATED IN BEARINGS.
2-DIMENSIONS ARE IN INCHES; TOLERANCES WHEN NOT SPECIFIED ARE ±.005 ON DECIMALS, ±1/64 ON FRACTIONS AND ±1° ON ANGLES.
3-THE DESIGN OF THIS UNIT SHALL PROVIDE FOR THE TERMINAL BLOCK TO REMAIN FIXED WHEN TERMINAL SCREWS AND TERMINAL LUGS ARE REMOVED.

FSC 5990

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TABLE I

RESOLVER, ELECTRICAL: TYPE 23R32N4A

PERFORMANCE REQUIREMENTS AND CHARACTERISTICS

	Requirements	Value	Unit	Tolerance
1.	Excitation winding	Stator		
2.	Rated frequency	400	Hz	±1%
3.	Input voltage		volts (rms)	±1%
	Maximum	130		
	Test	60		
	Minimum	1.0		
4.	Primary current	19	milliamps	nominal
5.	Primary power	0.08	watts	nominal
6.	D. C. resistance		ohms	±15%
	Rotor	170		
	Stator	26		
7.	Impedance		ohms	R±15%+jx±10%
	Z_{RO}	300+j3500		
	Z_{SO}	180+j3195		
	Z_{RS}	230+j255		
8.	Transformation ratio			nominal
	Output/input	0.990		±1%
9.	T. R. variation		%	
	Due to voltage	0.05		maximum
10.	Equality of transformation ratios		% spread	
	Output/input	0.2		
11.	Phase shift		degrees	
	Output/input	0.90		±0.5
12.	Phase shift variation		degrees	
	Due to voltage	±0.2		
13.	Interaxis error	±2.5	minutes	
14.	Functional error	±0.1	%	
15.	Null voltage		mV(rms) output	maximum
	Fundamental	21		
	Total	31.8		
16.	Shift of resolver zero		minutes	
	With voltage	±2.0		
	With frequency	±1.0		
17.	Harmonic distortion	0.1	%	maximum
18.	Temperature rise	20	°C	maximum
19.	Friction torque	0.2	oz. - in.	maximum
	@ -55°C	0.4		maximum

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TABLE I (Continued)

	Requirements	Value	Unit	Tolerance
20.	Radial play	0.0010 (for reversal of 1/2 lb load)	inches	maximum
21.	End play	0.0050 (for reversal of 1 lb load)	inches	maximum

Custodian:
Army - MU

Review activities:
Army - EL, AV

User activities:
Army - ME

Preparing activity:
Army - MU

(Project 5990-A016)

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